



ISTN7MP MOBILE AND PERSONAL INFORMATION SYSTEMS HONOURS MODULE – SEMESTER 1 - 2026

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MOBILE AND PERSONAL INFORMATION SYSTEMS

- PART I : MOBILE COMPUTING
- PART II : MOBILE AND PERSONAL INFORMATION SYSTEMS
- PART III : ANDROID MOBILE APP DEVELOPMENT

PART III

MOBILE OS TRENDS & ARCHITECTURE
AND
ANDROID APPLICATION DEVELOPMENT

MOBILE APPLICATION DEVELOPMENT

ANDROID APP DEVELOPMENT

APP DEMO – 4

DURBAN ATTRACTIONS APP

ACTIVITY LIFE CYCLE

- EACH ACTIVITY HAS A LIFE CYCLE, WHICH IS THE SERIES OF ACTIONS FROM THE BEGINNING OF AN ACTIVITY TO ITS END
- ACTIONS THAT OCCUR DURING THE LIFE CYCLE PROVIDE WAYS TO MANAGE HOW USER INTERACT WITH YOUR APP.
- EACH ACTIVITY BEGINS WITH AN ONCREATE() METHOD. THE ONCREATE() METHOD INITIALIZES THE USER INTERFACE WITH AN XML LAYOUT; THE LIFE OF THE ACTIVITY IS STARTED
- AS IN ANY LIFE CYCLE THE OTHER EXTREME OF BIRTH IS DEATH. IN THIS CASE, AN ONDESTROY() METHOD IS THE END OF THE ACTIVITY

ACTIVITY LIFE CYCLE (CONTD...)

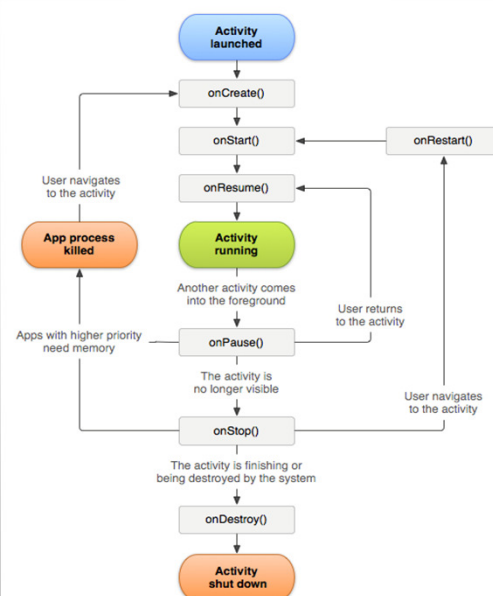
- THE ONCREATE() METHOD SETS UP ALL THE RESOURCES REQUIRED TO PERFORM THE ACTIVITY AND ONDESTROY() RELEASES THOSE SAME RESOURCES TO FREE UP MEMORY ON YOUR MOBILE DEVICE
- THERE ARE SEVERAL STAGES OF ACTIVITY IN BETWEEN ONCREATE() AND ONDESTROY() – SUCH AS ONRESTART(), ONSTART(), ONRESUME(), ONPAUSE(), AND ONSTOP()
- EACH OF THESE METHODS CHANGES THE STATE OF THE ACTIVITY
- THE DIFFERENT STATES OF THE ACTIVITY DETERMINE WHETHER THE ACTIVITY IS ACTIVE/STARTED/CREATED, PAUSED, STOPPED OR DEAD/DESTROYED

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ACTIVITY LIFE CYCLE (CONTD...)



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ACTIVITY LIFE CYCLE (CONTD...)

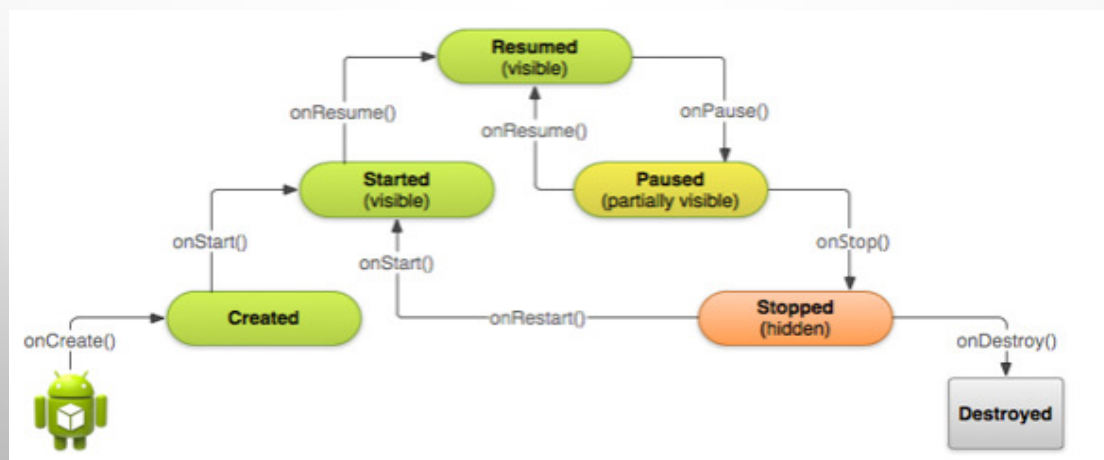
- **ONCREATE()** – BASIC APPLICATION STARTUP LOGIC, ALLOCATES REQUIRED RESOURCES TO THE ACTIVITY
- **ONSTART()** – MAKES ACTIVITY VISIBLE TO THE USER
- **ONRESUME()** – THE STATE IN WHICH ACTIVITY INTERACTS WITH THE USER
- **ONPAUSE()** – THOUGH STILL VISIBLE TO THE USER THE ACTIVITY IS NO LONGER IN THE FOREGROUND
- **ONSTOP()** – WHEN ACTIVITY IS NO LONGER VISIBLE (HIDDEN) TO THE USER IT ENTERS INTO ONSTOP STATE WHEN THE ACTIVITY IS FINISHED RUNNING AND ABOUT TO BE TERMINATED
- **ONRESTART()** – SWITCH THE ACTIVITY FROM ONSTOP TO ONSTART, MAKES THE ACTIVITY VISIBLE TO THE USER
- **ONDESTROY()** – WHEN STOPPED ACTIVITY CAN BE TERMINATED, RELEASES THE ALLOCATED RESOURCES FROM THE ACTIVITY

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ACTIVITY LIFE CYCLE (CONTD...)



DURBAN ATTRACTIONS APP

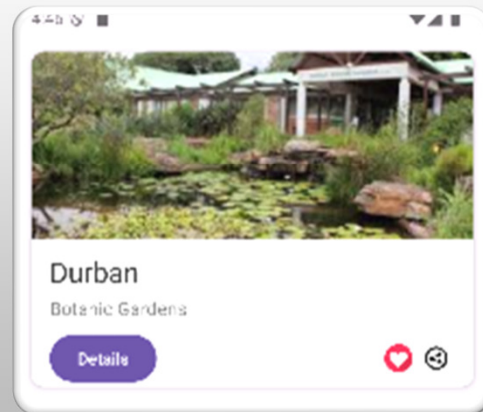
- THIS APP LISTS A LIST OF ATTRACTIONS THAT CAN BE VISITED BY A TOURIST IN DURBAN, SOUTH AFRICA
- THIS APP MAINLY FOCUS ON USER INTERFACE DESIGN SPECIFICALLY THROUGH XML CODE BY DIRECTLY CAPTURING XML CODE IN THE “ACTIVITY_MAIN.XML”
- THIS APP USES SCROLL VIEW TO LIST SERIES OF ATTRACTIONS. IN THIS CASE ITS FIVE ATTRACTIONS.
- SCROLL VIEW CAN BE ACHIEVED THROUGH – CARD VIEW AND RECYCLER VIEW
- OLDER VERSIONS OF ANDROID SUPPORTED LIST VIEW

WHAT'S NEW IN THIS APP

- DIRECT UI DESIGN USING XML
- FAMILIARITY ON LAYOUTS (SPECIFICALLY LINEAR LAYOUT)
- CARD VIEW
- SCROLL VIEW

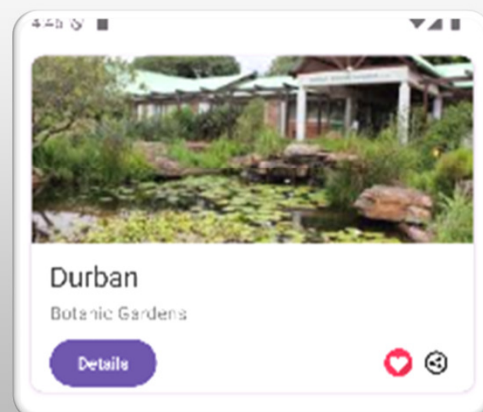
DURBAN ATTRACTIONS APP – CARD VIEW

- CARD VIEW – UI COMPONENT IN ANDROID – PROVIDES A CONTAINER WITH A SHADOW, ROUNDED CORNERS AND ELEVATION
- CARD VIEW – PART OF THE “android.cardview.widget.CardView” library
- “android.cardview.widget.CardView” is used to display in a structured, visually appealing format
- CARD VIEW IS COMPATIBLE WITH RECYCLER VIEW



DURBAN ATTRACTIONS APP – ATTRACTIONS

- BOTANIC GARDENS
- MOSES MABHIDA STATDIUM
- ANCIENT MUSEUM
- USHAKA MARINE WORLD
- GATEWAY MALL



DURBAN ATTRACTIONS APP – DEMO PLAN

- FIRST DEMONSTRATE THE CARD VIEW APP CONTAINING ONE ATTRACTION
(REFER “05 FOR ONE ATTRACTION.TXT” FILE)
- NEXT DEMONSTRATE THE FIVE ATTRACTIONS APP CONTAINING FIVE ATTRACTIONS
(REFER “05 FOR FIVE ATTRACTIONS.TXT” FILE)
- FINALLY DEMONSTRATE THE DURBAN ATTRACTIONS APP CONTAINING FIVE ATTRACTIONS IN SCROLL VIEW
(REFER “05 FINAL FOR SCROLL VIEW.TXT” FILE)

DURBAN ATTRACTIONS APP – STEPS TO PERFORM

- CREATE NEW PROJECT
- DELETE THE DEFAULT EXISTING TEXTVIEW CONTROL IN THE “ACTIVITY_MAIN.XML”
- CHOOSE THE SPLIT VIEW FROM THE LAYOUT DESIGN MENU OPTION. THIS VIEW WILL ALLOW BOTH THE LAYOUT EDITOR AND THE XML EDITOR IN THE ANDROID PROJECT SCREEN
- CAPTURE ALL THE REQUIRED STRINGS INTO THE “STRINGS.XML” FILE IN THE RESOURCES FOLDER
- COPY ALL THE “.PNG” IMAGE FILES INTO THE DRAWABLE FOLDER IN THE RESOURCES FOLDER
- CAPTURE THE XML CODE FOR ONE ATTRACTION INTO THE “ACTIVITY_MAIN.XML” FILE
- COPY THE SAME CODE FOUR TIMES AND MAKE CHANGES TO THE APPROPRIATE PROPERTIES INTO THE “ACTIVITY_MAIN.XML” FILE. THIS WILL DISPLAY ALL FIVE ATTRACTIONS
- MODIFY THE “ACTIVITY_MAIN.XML” FILE TO SUPPORT SCROLL VIEW

DURBAN ATTRACTIONS APP – CAPTURING STRINGS

- CAPTURE ALL THE REQUIRED STRINGS INTO THE “STRINGS.XML” FILE IN THE RESOURCES FOLDER

Key	Value
txtCity	Durban
txtBg	Botanic Gardens
txtGw	Gateway Mall
txtMus	Ancient Museum
txtStad	Moses Mabhida Stadium
txtUsh	Ushaka Marine World
btnText	Details
description	Images of Attraction

DURBAN ATTRACTIONS APP – COPY IMAGE FILES

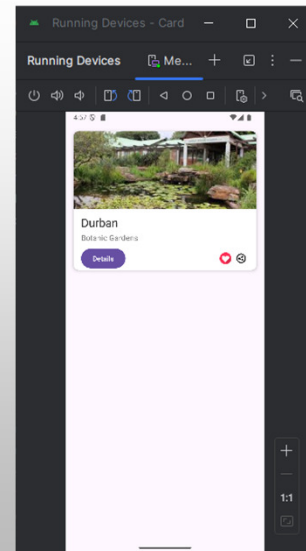
- COPY ALL THE “.PNG” IMAGE FILES INTO THE DRAWABLE FOLDER IN THE RESOURCES FOLDER

botanicg.png
gatewaym.png
mosesms.png
museumd.png
ushakamw.png
ic_launcher_like.png
ic_launcher_share.png

DURBAN ATTRACTIONS APP XML CODE FOR ONE ATTRACTION

CAPTURE THE XML CODE FOR ONE ATTRACTION INTO
THE "ACTIVITY_MAIN.XML" FILE

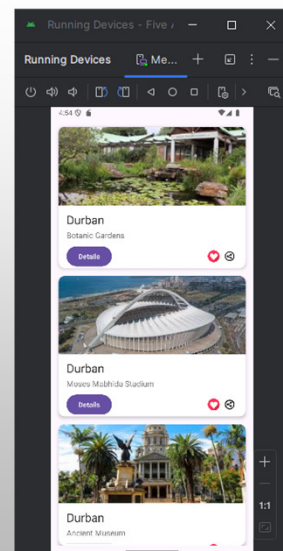
- REFER XML CODE IN "05 FOR ONE
ATTRACTION.TXT" FILE
- CAPTURE IT INTO THE "ACTIVITY_MAIN.XML" FILE



DURBAN ATTRACTIONS APP XML CODE FOR FIVE ATTRACTIONS

CAPTURE THE XML CODE FOR FIVE ATTRACTIONS
INTO THE "ACTIVITY_MAIN.XML" FILE

- REFER XML CODE IN "05 FOR FIVE
ATTRACTIONS.TXT" FILE
- CAPTURE IT INTO THE "ACTIVITY_MAIN.XML" FILE
- DISPLAYS THREE ATTRACTIONS BUT TWO ARE
HIDDEN, AS SCROLL VIEW IS NOT ENABLED



DURBAN ATTRACTIONS APP XML CODE FOR SCROLL VIEW

MODIFY THE "ACTIVITY_MAIN.XML" FILE TO SUPPORT SCROLL VIEW

- REFER XML CODE IN "05 FINAL FOR SCROLL VIEW.TXT" FILE
- CAPTURE IT INTO THE "ACTIVITY_MAIN.XML" FILE

ANDROID PRACTICE ACTIVITY 4(A)

1. PRACTICE THE SAME APP DEMONSTRATED DURING THE LECTURE SESSION
2. THE DURBAN ATTRACTIONS APP IN THE CURRENT FORM IS ONLY AT THE LEVEL OF UI DESIGN. MODIFY THE SAME APP ADDING FUNCTIONALITY. THE FOLLOWING ARE THE FEW SUGGESTIONS TO PROVIDE FUNCTIONALITY
 - a) ADD CODE FOR THE BUTTON. PROVIDE A VARIETY OF RESPONSES WHEN THE BUTTON OF EACH CARDVIEW ARE CLICKED. FOR EXAMPLE, NAVIGATING TO THE ATTRACTION'S WEBSITE, OR DISPLAYING ADDITIONAL DETAILS IN ANOTHER ACTIVITY. THE DETAILS MAY BE IN THE FORM OF TEXT OR IMAGES.
 - b) PROVIDE FUNCTIONALITY FOR THE LIKES AND SHARES. MAINTAIN A COUNT FOR THE NUMBER OF LIKES AND SHARES.

ANDROID PRACTICE ACTIVITY 4(B)

IMPLEMENT THE SAME APP (DURBAN ATTRACTIONS) USING RECYCLERVIEW.

ANDROID PRACTICE ACTIVITY 4(C)

EXTEND THE APP TO THE EXTENT OF SOUTH AFRICA. CALL IT SA ATTRACTIONS. MAKE USE OF SPINNER CONTROL IN THE MAIN ACTIVITY TO CHOOSE THE CITY FOR WHICH THE ATTRACTIONS NEED TO BE DISPLAYED IN THE TRAILING ACTIVITIES.

THANK YOU

END OF
MOBILE APP DEVELOPMENT: APP DEMO 4

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